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Section 5.52. Linear Measures

A. Application

A.1. General. – This code applies to any linear measure or measure of length, whether flexible or inflexible, permanently installed or portable.

A.2. Additional Code Requirements. – In addition to the requirements of this code, Linear Measures shall meet the requirements of Section 1.10. General Code.

S. Specifications

S.1.M. Units. – A linear measure may be in total length, and the total length may be subdivided in any or all of the following:

(a) centimeters and tenths of the centimeter;

(b) meters; and

(c) multiples of meters.

A 1-meter measure may be graduated, in addition, to show 0.1 m and multiples of 0.1 m subdivisions.

S.1. Units. – A linear measure may be in total length, and the total length may be subdivided in any or all of the following:

(a) inches and binary submultiples of the inch;

(b) feet;

(c) yards and multiples of yards.

A 1-yard measure may be graduated, in addition, to show $\frac{1}{8}$ yd and $\frac{1}{16}$ yd subdivisions. A flexible tape may be graduated in tenths or hundredths of a foot, or both tenths and hundredths of a foot. (Any other subdivisions are allowable only on measures of special purposes and when required for such purposes.)

S.2. Material.

S.2.1. Flexible Tape. – A flexible tape shall be made of metal.

S.2.2. End Measure. – If an end measure is made of material softer than brass, the ends of the measure shall be protected by brass (or other metal at least equally hard) securely attached.

S.3. Finish. – Measures shall be smoothly finished.

S.4. Design.

S.4.1. Rigid Measure. – A rigid measure shall be straight.

S.4.2. Folding Measure. – A folding measure shall open to a definite stop, and when so opened shall be straight.

S.5. Graduations.

S.5.1. General. – Graduations shall be perpendicular to the edge of the measure.

S.5.2. Width. – The width of the graduations on any measure shall not exceed one-half the width of the smallest graduated interval on the measure, and in no case shall be wider than 0.75 mm (0.03 in).

(Amended 1982)

T. Tolerances

T.1. For Measures Except Metal Tapes. – Maintenance tolerances in excess and in deficiency for measures except metal tapes shall be as shown in Table 1. Maintenance Tolerances, in Excess and in Deficiency, for Linear Measures Except Metal Tapes. Acceptance tolerances shall be one-half the maintenance tolerances.

<u>Table 1.</u> <u>Maintenance Tolerances, in Excess and in Deficiency, for Linear Measures Except Metal Tapes</u>	
<u>Nominal Interval from Zero</u>	<u>Tolerance</u>
<u>Feet</u>	<u>Inch</u>
<u>½ or less</u>	<u>1/64</u>
<u>1</u>	<u>1/32</u>
<u>2</u>	<u>1/16</u>
<u>3</u>	<u>3/32</u>
<u>4</u>	<u>1/8</u>
<u>5</u>	<u>5/32</u>
<u>6</u>	<u>3/16</u>

T.2. For Metal Tapes. – Maintenance and acceptance tolerances in excess and in deficiency for metal tapes shall be as shown in Table 2. Maintenance and Acceptance Tolerances, in Excess and in Deficiency, for Metal Tapes. Tapes of 10 m (25 ft) or over shall be tested at a tension resulting from a load of 5 kg (10 lb). Tapes less than 10 m (25 ft) shall be tested at a tension resulting from a load of 2.5 kg (5 lb). However, flexible metal tapes of 10 m (25 ft) or less that are not normally used under tension shall be tested with no tension applied. All tapes shall be supported throughout on a horizontal flat surface whenever tested.

(Amended 1972)

Table 2.	
Maintenance and Acceptance Tolerances, in Excess and in Deficiency, for Metal Tapes	
<u>Nominal Interval from Zero</u>	<u>Tolerance</u>
<u>Feet</u>	<u>Inch</u>
<u>6 or less</u>	<u>$\frac{1}{32}$</u>
<u>7 to 30, inclusive</u>	<u>$\frac{1}{16}$</u>
<u>31 to 55, inclusive</u>	<u>$\frac{1}{8}$</u>
<u>56 to 80, inclusive</u>	<u>$\frac{3}{16}$</u>
<u>81 to 100, inclusive</u>	<u>$\frac{1}{4}$</u>

~~Electrify Program~~ ~~Qualified Contractor Form Template~~

Overview: The goal of this Qualified Contractor Form template is to provide local governments and community organizations with easily modifiable content to integrate into an online qualified contractor form. Interested heat pump contractors can use this form to apply to be listed on the residential heat pump program's qualified contract list.

Instructions: You may use or modify this template in whatever way is most helpful. RMI does not expect credit or citation for any of this material.

Follow the proposed directions below to utilize this template:

1. Utilize Microsoft Word's replace all function (Ctrl/Command + shift + h) to make the following replacements:
 - a. Replace "[PROGRAM NAME]" with the name of your program. For example, "Electrify Ann Arbor".



- b. ~~Replace “[PROGRAM COORDINATOR]” with the organization that is coordinating or managing the program. For example, “The City of Ann Arbor.”~~
- c. ~~Replace “[JURISDICTION]” with the city, county, or neighborhood where your program is focused. For example, “Ann Arbor.”~~
- 2. ~~Replace all [yellow highlighted bracketed text] with the appropriate language for your project and local context.~~
- 3. ~~If the Qualified Contractor form is being released by the city directly, carefully review the entire Qualified Contractor form with your local government or organization’s attorney, procurement officer, and other relevant staff.~~
- 4. ~~To set up a Google Form, please follow the Google Form instructions found [here](#). To set up a Microsoft Form, please follow the Microsoft Form instructions [here](#).~~

Sources: This template was created using the best practices our team identified in various Electrify Programs in 2022 and 2023 and RMI’s internal expertise.

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Qualified Contractor Form Overview

Program Overview: [PROGRAM NAME] invites local air source heat pump (ASHP) and heat pump water heater (HPWH) contractors serving [JURISDICTION] to apply for participation in [PROGRAM NAME] as a Qualified Heat Pump Contractor (QHPC). The goals of [PROGRAM NAME] include [list program goals].

Program Partner Requirements: [PROGRAM NAME]’s partners will perform community outreach and communications to [JURISDICTION] residents to encourage them to sign up for [PROGRAM NAME]. QHPCs will [automatically receive all generated sign up leads OR automatically receive a percentage of generated sign up leads OR be recommended to residents to reach out to directly].

Qualified Contractor Requirements: QHPCs will be responsible for responding to residents that sign up for [PROGRAM NAME] within [two] business days, collecting the necessary information to provide residents an ASHP and/or HPWH quote, and supporting residents through the installation process should they sign a contract. QHPCs will be required to share final contract details, including [system technology, AHRI, price with rebates], with [PROGRAM NAME] administrators to privately track ASHP and HPWH installation and cost trends in [JURISDICTION].

Qualified Contractor Submission and Evaluation: Please fill out this form and submit by end of business on [DATE] to apply to our Qualified Heat Pump Contractor list. You will receive a response within [X business days]. If you have questions or concerns, please email [PROGRAM COORDINATOR EMAIL]. [PROGRAM NAME] reserves the right to remove QHPCs from the list if they do not meet minimum requirements of [minimum customer response time, minimum annual installation rates, customer satisfaction and reviews, and heat pump performance if monitoring systems available] and do not improve upon formal notice of warning from [PROGRAM NAME].—

Qualified Contractor Form Questions

1. —What is your company name?
2. —How many years has your company been in business?
3. —Who is your main point of contact's name, email, and phone number?
4. —What is your broader service territory coverage?
5. —What languages does your firm have working professional proficiency in?
6. —How many of the following technologies have you installed in the past twelve months? In the last three years? (Put in zero if you do not install this technology). A) Ducted ASHPs, B) Ductless ASHPs, C) HPWHs
7. —What ASHP or HPWH brands or manufacturer products do you supply?
8. —List all ASHP or HPWH trainings you have taken in the last three years.
9. —Attach two ASHP and/or two HPWH quotes from within the last six months.
10. —Provide two customer names and contact information for referral.
11. —What financial offerings do you offer?
12. —What community building efforts has your firm participated in?
13. —Explain in a few sentences why you are interested in [PROGRAM NAME].
14. —What are any questions or concerns you have?